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CONTROL DEVICE, FOREIGN SUBSTANCE REMOVAL DEVICE, AND CONTROL METHOD

Abstract

A control device of the present disclosure is used in a foreign substance removal device including a removal section configured to pick up and remove a foreign substance from a waste material conveyed by a conveyance device, a dust-proof section configured to perform dust-proofing related to maintenance of the foreign substance removal device with respect to powder dust derived from the waste material, and an imaging section located upstream of the removal section and configured to capture an image of the waste material. The control device executes one or more of control processing of controlling the dust-proof section based on information regarding an amount of waste material obtained from an image captured by the imaging section, and output processing of outputting information regarding the maintenance of the foreign substance removal device based on the information regarding the amount of waste material obtained from the image captured by the imaging section.

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Background/Summary

TECHNICAL FIELD

[0001] The present description discloses a control device, a foreign substance removal device, and a control method.

BACKGROUND ART

[0002] Conventionally, a processing device containing combustible powder dust has been proposed in which a combustible material such as waste plastic as a waste material is crushed by a crusher and burned in a heat process furnace (for example, refer to Patent Literature 1). In this processing device, the height of a processed object is detected using a height sensor, and the supply amount of the water vapor is controlled according to the height, so that the occurrence of powder dust explosion can be prevented in advance.

PATENT LITERATURE

[0003] Patent Literature 1: JP-A-2007-46797

SUMMARY OF THE INVENTION

Technical Problem

[0004] However, Patent Literature 1 does not consider maintenance of the device. For example, it is desirable that appropriate maintenance can be executed when removing a foreign substance from the waste material.

[0005] The present disclosure has been made to solve such a problem, and a main object thereof is to provide a control device, a foreign substance removal device, and a control method capable of executing more appropriate maintenance when removing a foreign substance from a waste material.

[0006] The present disclosure employs the following means in order to achieve the main object described above.

[0007] A control device disclosed in the present description is a control device used in a foreign substance removal device including a removal section configured to pick up and remove a foreign substance from a waste material conveyed by a conveyance device, a dust-proof section configured to perform dust-proofing related to maintenance of the foreign substance removal device with respect to powder dust derived from the waste material, and an imaging section located upstream of the removal section and configured to capture an image of the waste material, the control device including [0008] a control section configured to execute one or more of control processing of controlling the dust-proof section based on information regarding an amount of waste material obtained from the captured image captured by the imaging section, and output processing of outputting information regarding the maintenance of the foreign substance removal device based on the information regarding the amount of waste material obtained from the captured image captured by the imaging section.

[0009] In this control device, the control processing of controlling the dust-proof section that performs dust-proofing related to the maintenance of the foreign substance removal device is executed based on the information regarding the amount of waste material obtained from the captured image obtained by capturing the image of the waste material. Alternatively, the control device executes the output processing of outputting the information regarding the maintenance of the foreign substance removal device based on the information regarding the amount of waste material obtained from the captured image obtained by capturing the image of the waste material. In this control device, since the information on control and maintenance of the dust-proof section is output according to an increase or decrease in the amount of waste material, it is possible to

execute more appropriate maintenance of the device when removing the foreign substance from the waste material.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. **1** is a diagram illustrating an example of a configuration of recycling system **10**.

[0011] FIG. **2** is a view illustrating an example of a schematic configuration of foreign substance removal device **30**.

[0012] FIG. **3** is a perspective view illustrating an example of a schematic configuration of removal section **34**.

[0013] FIG. **4** is a diagram illustrating an example of information stored in storage section **52**.

[0014] FIG. **5** is a graph of correspondence relationship information **55** including a correspondence relationship between a waste material image and the amount of powder dust.

[0015] FIG. **6** is a graph of correspondence relationship information **55** including a correspondence relationship between the amount of powder dust and the dust-proof control amount.

[0016] FIG. **7** is a diagram illustrating an example of captured images **81** and **82** of recognition section **31**.

[0017] FIG. **8** is a flowchart illustrating an example of a foreign substance removal processing routine.

[0018] FIG. **9** is a flowchart illustrating an example of a dust-proof control output processing routine.

[0019] FIG. **10** is a graph of correspondence relationship information including a correspondence relationship between a waste material image and the amount of waste materials.

[0020] FIG. **11** is a graph of correspondence relationship information including a correspondence relationship between the amount of waste materials and the dust-proof control amount.

[0021] FIG. **12** is a flowchart illustrating an example of another dust-proof control output processing routine.

DESCRIPTION OF EMBODIMENTS

[0022] An embodiment of the present disclosure will be described with reference to the accompanying drawings. FIG. **1** is a diagram illustrating an example of a configuration of recycling system **10** for recycling waste material. FIG. **2** is a view illustrating an example of a schematic configuration of foreign substance removal device **30**. FIG. **3** is a perspective view illustrating an example of a schematic configuration of removal section **34**. FIG. **4** is a diagram illustrating an example of accumulated information **53**, threshold value information **54**, and correspondence relationship information **55** stored in storage section **52**. FIG. **5** is a diagram illustrating an example of correspondence relationship information **55** including a correspondence relationship between a waste material image and the amount of powder dust in a captured image. FIG. **6** is a diagram illustrating an example of correspondence relationship information **55** including the correspondence relationship between the amount of powder dust and the dust-proof control amount. FIG. **7** is a diagram illustrating an example of captured images **81** and **82** of recognition section **31**. In the present embodiment, a left-right direction, a front-rear direction, and an up-down direction are as illustrated in FIGS. **1** to **3**, and **7**. In addition, in the present embodiment, a direction where waste material **12** is conveyed is referred to as conveyance direction D.

[0023] Waste material **12** processed by recycling system **10** is a mixture obtained by mixing target object **13**, which is a target for recycling such as stone, sand, and concrete, with foreign substance **14** including paper, resin, wood, metal, or the like. As illustrated in FIG. **1**, recycling system **10** includes primary crusher **15**, primary magnetic separator **16**, screen machine **17**, foreign substance removal device **30**, secondary crusher **18**, secondary magnetic separator **19**, and conveyance

devices **20** to **25**.

[0024] Primary crusher **15** is a device that primarily crushes waste material **12** as a raw material. Primary crusher **15** crushes waste material **12** so as to have a predetermined primary size or less (for example, 40 cm or less), for example. Primary magnetic separator **16** is a device that removes the foreign substance of magnetic material contained in waste material **12** by magnetic force. Screen machine **17** is, for example, a device that separates waste material **12** having a primary size or more and waste material **12** having a size less than the primary size by passing waste material **12** over a mesh. Secondary crusher **18** is a device that secondarily crushes waste material **12** to a size smaller than that of primary crusher **15**. Secondary crusher **18** crushes waste material **12** so as to have a predetermined secondary size or less (for example, 10 cm or less). Secondary magnetic separator **19** is a device that removes, from waste material **12**, the magnetic material not removed by primary magnetic separator **16** and foreign substance removal device **30**.

[0025] Conveyance devices **20** to **25** as conveyance lines are devices that place waste material **12** on a conveyance surface and convey waste material **12** along conveyance direction D, and are configured as, for example, belt conveyors. Conveyance devices **20** to **25** may have configurations other than the belt conveyor as long as waste material **12** is conveyed.

[0026] Foreign substance removal device **30** is a device that is placed on the conveyance surface of conveyance device **22**, conveys waste material **12** in the conveyance line, and removes foreign substance **14**. As illustrated in FIG. 2, foreign substance removal device **30** includes recognition section **31**, removal section **34**, control device **50**, and dust-proof section **60**. Conveyance device **22** places waste material **12** on the conveyance surface and conveys waste material **12** at a predetermined conveyance speed.

[0027] Recognition section **31** is disposed upstream of removal section **34** in conveyance direction D. Recognition section **31** includes imaging section **32** that captures an image of foreign substance **14** contained in waste material **12** conveyed by conveyance section **22**, and detection section **33** that detects the height of waste material **12**. Imaging section **32** is configured as, for example, a camera that captures an image of waste material **12** placed on conveyance surface **30** of conveyance device **22** and conveyed. Imaging section **32** is held above the conveyance surface of conveyance device **22** by a holding member provided so as to straddle conveyance device **22** in the left-right direction. Imaging section **32** captures an image of a predetermined range including waste material **12** from above the conveyance surface of conveyance device **22**, and outputs image data to control device **50** of foreign substance removal device **30**. As illustrated in FIG. 7, imaging section **32** captures captured images **81** and **82** including target object **13** and foreign substance **14**. Control device **50** acquires widths W1 and W2 of waste material **12** from the captured image, and acquires the amount of waste materials **12**, the amount of powder dust from waste material **12**, and the like. Detection section **33** detects the height of waste material **12** on conveyance device **22**. For example, detection section **33** may be provided upstream of removal section **34** in conveyance direction D, and may be configured as a stereo camera including two cameras. Detection section **33** is held above conveyance surface of conveyance device **22** by the holding member provided so as to straddle conveyance device **22** in the left-right direction. Detection section **33** outputs information regarding the detected height of waste material **12** to control device **50**. As illustrated in FIG. 2, recognition section **31** may include dust-proof cover **48** that covers imaging section **32**. Foreign substance removal device **30** including detection section **33** can perform conveyance control of waste material **12** using information in the height direction of waste material **12**. Foreign substance removal device **30** may omit detection section **33** or may omit the use of information on the height of waste material **12**.

[0028] Removal section **34** is a device that removes foreign substance **14** from waste material **12** conveyed by conveyance device **22**, and is configured as, for example, an XY robot. Removal section **34** removes foreign substance **14** present in waste material **12** based on the captured image captured by imaging section **32**. Removal section **34** is provided downstream of recognition section

31 in conveyance direction **D**. Removal section **34** includes X-axis slider **35**, Y-axis slider **40**, lifting and lowering section **41** (refer to FIG. 5), pickup section **46**, support portion **57**, removal head **58**, and dust box **59**. Removal section **34** discards pickup foreign substance **14** into dust box **59** disposed on the side of conveyance device **22**. Removal section **34** may have another configuration such as an articulated arm robot.

[0029] X-axis slider **35** includes a guide rail installed along a direction orthogonal to conveyance direction **D** of conveyance device **22**, a slider that moves along the guide rail, and drive mechanism **36** that drives the slider. The guide rail is fixed to a slider of Y-axis slider **40** installed on conveyance section **22** along conveyance direction **D**. Removal head **58** provided with lifting and lowering section **41** is fixed to the slider. Drive mechanism **36** may be, for example, a linear motion mechanism using a ball screw including screw shaft **37**, nut **38**, and motor **39**. Drive mechanism **36** is not particularly limited as long as it is a linear motion mechanism, and may be a linear motor or the like. In X-axis slider **35**, removal head **58** is moved along the X-axis direction by drive mechanism **36**.

[0030] Y-axis slider **40** includes a guide rail installed along conveyance direction **D**, a slider that moves along the guide rail, and a drive mechanism that drives the slider. The guide rail is fixed to support portion **57** installed on conveyance section **22** along conveyance direction **D**. X-axis slider **35** is disposed on the slider. The drive mechanism may be, for example, a linear motion mechanism using a ball screw including a screw shaft, a nut, and a motor. The drive mechanism has the same configuration as that of drive mechanism **36**, and the description thereof is omitted. In Y-axis slider **40**, X-axis slider **35** is moved along the Y-axis direction by the drive mechanism.

[0031] Lifting and lowering section **41** is fixed to removal head **58** disposed on X-axis slider **35**. Lifting and lowering section **41** includes drive mechanism **42** that lifts and lowers pickup section **46** along the Z-axis. Drive mechanism **42** may be, for example, a linear motion mechanism using a ball screw including screw shaft **43**, nut **44**, and motor **45**. Drive mechanism **42** is not particularly limited as long as it is a linear motion mechanism, and may be a linear motor or the like. In lifting and lowering section **41**, pickup section **46** is lifted and lowered along the Z-axis direction by drive mechanism **42**.

[0032] Pickup section **46** includes multiple gripping claws. Pickup section **46** grips and releases foreign substance **14** by the opening and closing operation of the gripping claws, and picks up and removes foreign substance **14** from waste material **12**. Pickup section **46** is opened and closed by a grip drive section (not illustrated). Pickup section **46** is moved in the X, Y, and Z directions within a predetermined range of conveyance section **22** by X-axis slider **35**, Y-axis slider **40**, and lifting and lowering section **41**.

[0033] Dust-proof section **60** is a unit that performs dust-proofing related to maintenance of foreign substance removal device **30** on powder dust derived from waste material **12**. Here, the term “dust-proofing” includes, for example, preventing powder dust from entering the inside of the device and removing the powder dust outside the device in a case where the powder dust enters the inside of the device. Dust-proof section **60** includes, for example, gas circulation section **61** for circulating gas for removing powder dust from waste material **12**, liquid circulation section **65** for circulating liquid for removing powder dust from waste material **12**, and filter section **75** for filtering powder dust from waste material **12**. Gas circulation section **61** performs dust-proofing by circulating gas in recognition section **31**, for example. Examples of the gas include air. Specifically, for example, gas circulation section **61** may blow a gas onto the lens surface of imaging section **32** to suppress the adhesion of powder dust. In this case, gas circulation section **61** includes air supply section **62**, supply pipe **63**, and discharge port **64**. Air supply section **62** may be a compressor that discharges compressed air. Supply pipe **63** is a tubular member that circulates gas from gas circulation section **61** to imaging section **32**. Discharge port **64** is an opening portion opened toward the lens surface of imaging section **32**. In gas circulation section **61**, cleaning of air supply section **62**, supply pipe **63**, and the like is executed depending on usage conditions including the amount of powder dust

and the like, as maintenance. Alternatively, for example, gas circulation section **61** may prevent powder dust from entering dust-proof cover **48** by circulating gas in dust-proof cover **48** of recognition section **31**. In this case, in gas circulation section **61**, the adjustment of the flow rate of the gas to be circulated is executed depending on usage conditions including the amount of powder dust and the like, as maintenance.

[0034] Liquid circulation section **65** is, for example, a device that washes away powder dust entering a movable section such as X-axis slider **35**, Y-axis slider **40**, and lifting and lowering section **41** with liquid. Examples of the liquid include oil. Liquid circulation section **65** includes slider dust-proof section **66** disposed on X-axis slider **35** and Y-axis slider **40**, and lifting and lowering dust-proof section **70** disposed on lifting and lowering section **41**. Slider dust-proof section **66** includes cleaning liquid tube **67** and circulation pump **68**. Cleaning liquid tube **67** is a tubular member through which a liquid circulates, and is connected to both end portions of drive mechanism **36**. Circulation pump **68** is a liquid feed pump, and circulates the liquid into drive mechanism **36** via cleaning liquid tube **67**. Lifting and lowering dust-proof section **70** includes cleaning liquid tube **71** and circulation pump **72**. Cleaning liquid tube **71** is a tubular member through which a liquid circulates, and is connected to both end portions of drive mechanism **42**. Circulation pump **72** is a liquid feed pump, and circulates the liquid into drive mechanism **42** via cleaning liquid tube **71**. A filter (not illustrated) is disposed inside liquid circulation section **65**. The filter separates foreign substances such as powder dust. In liquid circulation section **65**, cleaning, filter exchange, and the like of cleaning liquid tubes **67** and **71**, circulation pumps **68** and **72**, and the like are executed depending on usage conditions including the amount of powder dust and the like, as maintenance. In filter section **75**, filter **76** is detachably disposed outside cooling fan **56** of control device **50**. Filter section **75** does not include a drive section that is controlled according to the amount of powder dust, but filter **76** is exchanged depending on usage conditions including the amount of powder dust and the like, as maintenance.

[0035] Control device **50** is a device that controls entire foreign substance removal device **30**, and includes control section **51**, storage section **52**, and cooling fan **56**. Control section **51** is configured as a microprocessor centered on the CPU, and controls the entire device. Control section **51** outputs control signals to recognition section **31**, removal section **34**, and dust-proof section **60**, and inputs signals from recognition section **31**, removal section **34**, and dust-proof section **60**. Control section **51** executes one or more of the control processing of controlling dust-proof section **60** based on information regarding the amount of waste materials **12** obtained from the captured image captured by imaging section **32** and the output processing of outputting information regarding the maintenance of each configuration of foreign substance removal device **30** based on the information regarding the amount of waste materials **12** obtained from the captured image captured by imaging section **32**. For example, control section **51** acquires the amount of powder dust from waste material **12** as the information regarding the amount of waste materials **12** from the captured image captured by imaging section **32**. Control section **51** may acquire the amount of waste materials as the information regarding the amount of waste materials **12**. Storage section **52** is a storage medium for storing information and includes, for example, an HDD or a flash memory. Cooling fan **56** performs waste heat for operating control section **51**, and is driven to a stepwise air blowing amount according to a load of control section **51**.

[0036] As illustrated in FIG. **4**, storage section **52** stores various control programs in addition to accumulated information **53**, threshold value information **54**, correspondence relationship information **55**, and the like. Accumulated information **53** is information in which the last maintenance execution date and time and the accumulated operation amount accumulated from the maintenance execution date and time are associated with each configuration of dust-proof section **60**. Examples of the configuration of dust-proof section **60** include gas circulation section **61** for performing dust-proofing for imaging section **32**, liquid circulation section **65** for performing dust-proofing for X-axis slider **35**, Y-axis slider **40**, and lifting and lowering section **41**, and filter section

75 for performing dust-proofing for control device **50**. The accumulated operation amount indicates a value obtained by accumulating the operation amount in which the load is added according to the amount of powder dust from waste material **12**. Threshold value information **54** is information in which an advance notice threshold value and a maintenance execution threshold value are associated with each configuration of dust-proof section **60**. The advance notice threshold value indicates entering a preparation period for maintenance execution based on the accumulated operation amount. The maintenance execution threshold value indicates entering the period for actually executing maintenance.

[0037] Correspondence relationship information **55** is information used when the amount of powder dust is acquired from the information on the waste material included in the captured image captured by imaging section **32** and the control amount of dust-proof section **60** is acquired from the amount of powder dust. Correspondence relationship information **55** can obtain the amount of powder dust from waste material **12** by using at least width W substantially orthogonal to conveyance direction D of waste material **12** included in the captured image. Correspondence relationship information **55** includes a correspondence relationship in which width W of waste material **12** in the captured image illustrated in FIG. **5** is associated with the amount of powder dust, and a correspondence relationship in which the amount of powder dust illustrated in FIG. **6** is associated with the control amount of dust-proof section **60**. Correspondence relationship information **55** has, for example, a correspondence relationship in which the larger width W of waste material **12** obtained from the captured image or the area of waste material **12** obtained from width W , the larger the amount of powder dust is obtained. In addition, in correspondence relationship information **55**, the control amount for controlling dust-proof section **60** is associated with a tendency that the smaller the amount of powder dust from waste material **12** obtained from the captured image, the more the operation of dust-proof section **60** is suppressed. In addition, in correspondence relationship information **55**, the control amount for controlling dust-proof section **60** is associated with a tendency that the larger the amount of powder dust from waste material **12** obtained from the captured image, the more the operation of dust-proof section **60** is increased. The term “with a tendency” refers to the correspondence relationship as a whole, but it means that it is caused to include a part of a portion of the correspondence relationship that is different from the correspondence relationship, such as a constant value or an inverse correspondence relationship. The correspondence relationship related to each configuration may be empirically obtained by experiments or the like. The correspondence relationship for obtaining the amount of powder dust may be a linear relationship as illustrated in FIG. **5A** or a curved relationship as illustrated in FIG. **5B**. In addition, the correspondence relationship for obtaining the control amount may be a proportional relationship as illustrated in FIG. **6A** with respect to a configuration in which the stepless control is possible, or may be a stepwise relationship as illustrated in FIG. **6B** with respect to a configuration in which only the stepwise control is possible.

[0038] Next, the operation of foreign substance removal device **30** configured as described above will be described. First, the processing of removing foreign substance **14** from waste material **12** will be described. FIG. **8** is a flowchart illustrating an example of a foreign substance removal processing routine executed by control section **51** of control device **50**. This routine is stored in storage section **52**, and is executed by control section **51** after foreign substance removal device **30** is activated. When this routine is started, control section **51** drives conveyance device **22** (**S100**), and determines whether it is an imaging timing (**S110**). The imaging timing may be set to a timing at which entire waste material **12** conveyed by conveyance device **22** is captured as a seamless still image. In a case where imaging section **32** images a moving image, this processing needs only to be omitted. When it is not the imaging timing in **S110**, control section **51** stands by as it is. When it is the imaging timing in **S110**, control section **51** causes imaging section **32** to capture the captured image and acquire the captured image to perform recognition processing of foreign substance **14** (**S120**). For example, control section **51** performs the processing of detecting a region different

from the luminance of target object **13** and setting the region as the region of foreign substance **14**, as recognition processing of foreign substance **14**. Next, control section **51** determines whether there is foreign substance **14** (S130), and when there is foreign substance **14**, executes foreign substance removal processing (S140). In the foreign substance removal processing, control section **51** moves pickup section **46** to the region where foreign substance **14** is recognized, grips foreign substance **14**, and then executes processing of moving foreign substance **14** to dust box **59**. After S140 or when there is no foreign substance **14** in S130, control section **51** determines whether the regeneration processing of waste material **12** is completed (S150). When the regeneration processing is not completed, control section **51** repeatedly executes the processing at S110 and the subsequent steps. Conversely, when the regeneration processing is completed in S150, control section **51** ends this routine. As described above, in foreign substance removal device **30**, the removal processing of foreign substance **14** is executed using the captured image obtained by imaging waste material **12** on conveyance device **22**.

[0039] Next, processing in which foreign substance removal device **30** notifies the execution of dust-proofing for each configuration and the maintenance of each configuration will be described. FIG. **9** is a flowchart illustrating an example of a dust-proof control output processing routine executed by control section **51**. This routine is stored in storage section **52**, and is executed by control section **51** in parallel with the foreign substance removal processing. Here, a case where the amount of powder dust derived from waste material **12** is acquired as the information regarding the amount of waste materials **12** and used for the processing will be mainly described. When this routine is started, control section **51** determines whether it is an imaging timing (S200) as in S110. When it is not the imaging timing, control section **51** stands by as it is. When it is the imaging timing, control section **51** acquires the captured image captured by imaging section **32**, and acquires the amount of powder dust from waste material **12** as the information regarding the amount of waste materials based on correspondence relationship information **55** (S210). Control section **51** acquires width **W** of waste material **12** on conveyance device **22** from the captured image at predetermined intervals along conveyance direction **D**. Control section **51** acquires the information regarding the amount of waste materials by using the information including at least width **W** of waste material **12**. Control section **51** may obtain the amount of powder dust from width **W** of waste material **12**. Alternatively, control section **51** may obtain the area of waste material **12** using width **W** and the predetermined intervals to obtain the amount of powder dust corresponding thereto. As illustrated in FIG. **7**, width **W** of waste material **12** is correlated with the amount of waste materials **12** present on conveyance device **22**. Control section **51** can obtain the amount of waste materials and the amount of powder dust according to width **W** using correspondence relationship information **55** empirically obtained as illustrated in FIGS. **5** and **6**. [0040] Next, control section **51** acquires the control amount of dust-proof section **60** corresponding to the acquired amount of powder dust using correspondence relationship information **55** (S230). Examples of the control amount include the amount of air supplied by air supply section **62**, the amount of liquid fed by circulation pump **68**, the amount of liquid fed by circulation pump **72**, and the like. Since control section **51** sets the control amount based on correspondence relationship information **55**, control section **51** sets the control amount of dust-proof section **60** with a tendency that the smaller the amount of powder dust from waste material **12** obtained from the captured image, the more the operation of dust-proof section **60** is suppressed. In addition, since control section **51** sets the control amount based on correspondence relationship information **55**, control section **51** sets the control amount of dust-proof section **60** with a tendency that the larger the amount of powder dust from waste material **12** obtained from the captured image, the more largely dust-proof section **60** is operated. Next, control section **51** drives and controls dust-proof section **60** using the set control amount (S240). Control section **51** acquires the operation amount corresponding to the control amount or the amount of powder dust, and updates accumulated information **53** with the acquired operation amount (S250). For example, control section **51** updates

the accumulated operation amounts according to the control amounts thereof for air supply section **62** and circulation pumps **68** and **72** that are directly driven. In addition, control section **51** updates the accumulated operation amount according to the amount of powder dust for filter section **75** which is not directly driven. For example, when the standard operation amount is 1, control section **51** may accumulate the operation amount according to the ratio to the standard control amount or the amount of powder dust. For example, when the accumulated operation amount is set to 10 in a case where the operation amount 1 continues for 10 minutes, control section **51** sets the accumulated operation amount to 5 in a case where the operation amount 0.5 continues for 10 minutes. In addition, in a case where the operation amount of 1.5 continues for 10 minutes, control section **51** sets the accumulated operation amount to 15. In dust-proof section **60**, when a load of the dust-proof processing is smaller, the accumulated operation amount also becomes a smaller value, and when a load of the dust-proof processing is larger, the accumulated operation amount also becomes a larger value.

[0041] When the accumulated operation amount of dust-proof section **60** is updated in **S250**, control section **51** determines whether the accumulated operation amount is equal to or greater than the advance notice threshold value set in threshold value information **54** (**S260**). Control section **51** executes the determination of the advance notice threshold value for each of the configurations of dust-proof section **60**. When the accumulated operation amount is equal to or greater than the advance notice threshold value, control section **51** determines whether the accumulated operation amount is equal to or greater than the maintenance execution threshold value (**S270**). When the accumulated operation amount is equal to or greater than the advance notice threshold value and less than the maintenance execution threshold value, control section **51** outputs a maintenance advance notice to the configuration of dust-proof section **60** (**S280**). For example, the maintenance advance notice may be one that displays and outputs information to the effect that maintenance is executed soon on a display section such as an operation panel. Worker M who has confirmed this notice can prepare to execute maintenance for the configuration of dust-proof section **60**. Conversely, when the accumulated operation amount is equal to or greater than the advance notice threshold value and is equal to or greater than the maintenance execution threshold value, control section **51** outputs the maintenance execution to the configuration of dust-proof section **60** (**S290**). Here, control section **51** may be a section that displays and outputs information to the effect that maintenance is executed on a display section such as an operation panel. In addition, control section **51** may output information on the exchange of filter **76** as the information regarding the maintenance. In a case where the maintenance is not executed and the device is left, control section **51** may stop the operation of foreign substance removal device **30** after a predetermined period has elapsed in order to prevent a failure of the device. Since the accumulated operation amount of dust-proof section **60** corresponds to the amount of powder dust from waste material **12**, control section **51** notifies the worker of the information regarding the maintenance in a longer period when the amount of powder dust from waste material **12** is small. Conversely, when the amount of powder dust from waste material **12** is large, control section **51** notifies the worker of the information regarding the maintenance in a shorter period.

[0042] After **S290**, after **S280**, or when the accumulated operation amount is less than the advance notice threshold value in **S260**, control section **51** determines whether to end the dust-proof control output routine (**S300**). For example, in a case where the regeneration processing of waste material **12** is completed, control section **51** determines to end the dust-proof control output routine. Alternatively, upon receiving an instruction to end the dust-proof control output routine, control section **51** determines to end the dust-proof control output routine. For example, the instruction to end the dust-proof control output routine may be input by the worker via the operation panel. When it is determined that the dust-proof control output routine is not ended, control section **51** executes the processing at **S200** and the subsequent steps. Conversely, when it is determined to end the dust-proof control output routine, control section **51** ends this routine. As described above, in foreign

substance removal device **30**, the information regarding the amount of waste materials **12** is obtained using the captured image obtained by capturing the image of waste material **12** on conveyance device **22**, the control amount of dust-proof section **60** is changed according to the information, the operation amount according to the control amount or the amount of powder dust is accumulated, and the maintenance is executed at a more appropriate timing.

[0043] Here, a correspondence relationship between the elements of the present embodiment and the elements of the present disclosure will be clarified. Foreign substance removal device **30** of the present embodiment is an example of the foreign substance removal device of the present disclosure, control device **50** is an example of the control device, imaging section **32** is an example of the imaging section, removal section **34** is an example of the removal section, dust-proof section **60** is an example of the dust-proof section, and control section **51** is an example of the control section. In addition, conveyance device **22** is an example of the conveyance device, waste material **12** is an example of the waste material, and foreign substance **14** is an example of the foreign substance. In the present embodiment, an example of the control method for the present disclosure is also clarified by describing the operation of foreign substance removal device **30**.

[0044] Foreign substance removal device **30** described above includes removal section **34** that picks up and removes foreign substance **14** from waste material **12** conveyed by conveyance device **22**, dust-proof section **60** that performs dust-proofing related to the maintenance of foreign substance removal device **30** with respect to powder dust derived from waste material **12**, and imaging section **32** located upstream of removal section **34** and that captures the image of waste material **12**. In addition, control device **50** includes control section **51** that executes one or more of control processing of controlling dust-proof section **60** based on the information regarding the amount of waste materials **12** obtained from the captured image captured by imaging section **32** and output processing of outputting the information regarding the maintenance of foreign substance removal device **30** based on the information regarding the amount of waste materials **12** obtained from the captured image captured by imaging section **32**. Control device **50** executes the control processing of controlling dust-proof section **60** that performs dust-proofing related to the maintenance of foreign substance removal device **30** based on the amount of powder dust as the information regarding the amount of waste materials **12** obtained from the captured image obtained by capturing the image of waste material **12**. Alternatively, control device **50** executes the output processing of outputting the information regarding the maintenance of foreign substance removal device **30** based on the amount of powder dust as the information regarding the amount of waste materials **12** obtained from the captured image obtained by capturing the image of waste material **12**. In control device **50**, since the information on the control and the maintenance of dust-proof section **60** is output according to the increase and decrease in the amount of waste materials **12**, it is possible to execute more appropriate maintenance of the device when removing foreign substance **14** from waste material **12**.

[0045] In addition, control section **51** acquires the amount of powder dust as the information regarding the amount of waste materials from the captured image, and executes the control processing and/or the output processing based on the acquired amount of powder dust. In control device **50**, since the amount of powder dust derived from waste material **12** can be obtained from the captured image, it is possible to more appropriately execute the maintenance of the device based on the amount of powder dust. Furthermore, control section **51** executes one or more of the control processing of controlling dust-proof section **60** based on the amount of powder dust and the output processing of outputting the information regarding the maintenance of foreign substance removal device **30** based on the amount of powder dust. In control device **50**, the control processing and the output processing can be more appropriately executed using the amount of powder dust obtained from the captured image. Furthermore, control section **51** obtains the amount of waste materials **12** and/or the amount of powder dust from waste material **12** based on information including at least width W of the waste material included in the captured image. In

control device **50**, the information regarding the amount of waste materials can be obtained by relatively easy processing using information including at least the width of waste material **12**. In addition, control section **51** may obtain the amount of waste materials and/or the amount of powder dust from the waste material from width **W** of the waste material included in the captured image, or may obtain the area by using width **W** of the waste material included in the captured image, and obtain the amount of waste materials and/or the amount of powder dust from the waste material from the area. In addition, control section **51** may obtain the information regarding the amount of waste materials by using correspondence relationship information **55** that empirically associates the information including at least width **W** of waste material **12** with the amount of waste materials and/or the amount of powder dust.

[0046] In addition, control section **51** executes the control processing of controlling dust-proof section **60** with a tendency that the smaller the amount of waste materials **12** obtained from the captured image and/or the amount of powder dust from waste material **12** obtained from the captured image, the more the operation of dust-proof section **60** is suppressed. In control device **50**, it is possible to further suppress an excessive operation of dust-proof section **60** or the like according to the amount of waste materials **12**, and it is possible to more appropriately execute maintenance of the device. Furthermore, control section **51** executes the output processing of outputting the information regarding the maintenance with a tendency that the smaller the amount of waste materials **12** obtained from the captured image and/or the amount of powder dust from waste material **12** obtained from the captured image, the longer an output interval of the information regarding the maintenance of foreign substance removal device **30**. In control device **50**, it is possible to further suppress the execution of excessive maintenance of dust-proof section **60** or the like according to the amount of waste materials **12**, and it is possible to more appropriately execute maintenance of the device. Furthermore, control section **51** executes the control processing of controlling dust-proof section **60** with a tendency that the larger the amount of waste materials **12** obtained from the captured image and/or the amount of powder dust from waste material **12** obtained from the captured image, the more the operation of dust-proof section **60** is increased. In control device **50**, it is possible to further suppress the operation shortage of dust-proof section **60** or the like according to the amount of waste materials **12**, and it is possible to more appropriately execute the dust-proof processing. In addition, control section **51** executes the output processing of outputting the information regarding the maintenance with a tendency that the larger the amount of waste materials **12** obtained from the captured image and/or the amount of powder dust from waste material **12** obtained from the captured image, the shorter the output interval of the information regarding the maintenance of foreign substance removal device **30**. In control device **50**, it is possible to further suppress insufficient maintenance of dust-proof section **60** or the like according to the amount of waste materials **12**, and it is possible to more appropriately execute maintenance of the device. In addition, in the output processing, control section **51** accumulates the operation amount of dust-proof section **60**, and outputs the information regarding the maintenance when the accumulated operation amount reaches a predetermined maintenance execution amount. In control device **50**, since the execution of the maintenance is notified according to the operation amount of dust-proof section **60**, it is possible to more appropriately execute the maintenance of the device.

[0047] In addition, dust-proof section **60** includes one or more of gas circulation section **61** for circulating a gas for removing powder dust from waste material **12**, liquid circulation section **65** for circulating a liquid for removing powder dust from waste material **12**, and filter section **65** for filtering the powder dust from waste material **12**. In control device **50**, the circulation of gas or liquid, or filter **76** can remove powder dust. Furthermore, control device **50** can more appropriately control the operation thereof. Furthermore, removal section **34** removes foreign substance **14** present in waste material **12** based on the captured image captured by imaging section **32**. In control device **50**, the amount of waste materials **12** can be obtained using the foreign substance

removal image captured by imaging section 32. Accordingly, it is possible to execute more appropriate maintenance of the device while further suppressing complications of the configuration without requiring other detection sections or the like.

[0048] In addition, foreign substance removal device 30 includes removal section 34 that picks up and removes foreign substance 14 from waste material 12 conveyed by conveyance device 22, dust-proof section 60 that performs dust-proofing related to the maintenance of foreign substance removal device 30 with respect to powder dust derived from waste material 12, imaging section 32 located upstream of removal section 34 and that captures the image of waste material 12, and the above-described control device 50. Since foreign substance removal device 30 includes control device 50 described above, it is possible to more appropriately execute maintenance of the device.

[0049] In addition, in dust-proof section 60, excessive operation can be further suppressed, and thus consumption of air, oil, and the like can be further suppressed. Furthermore, consumable items such as oil and a filter of dust-proof section 60 can be exchanged at an appropriate timing.

Furthermore, in foreign substance removal device 30, since imaging section 32 that detects foreign substance 14 can be used, it is possible to obtain the information regarding the amount of waste materials 12 while further suppressing the addition of a configuration and further suppressing the additional cost.

[0050] It is needless to say that the present disclosure is not limited in any way to the above-described embodiments, and the present disclosure can be embodied in various aspects as long as the aspects fall within the technical scope of the present disclosure.

[0051] For example, in the embodiment described above, the amount of powder dust is acquired as the information regarding the amount of waste materials from the captured image, and the control processing and the output processing are executed based on the acquired amount of powder dust; however, the configuration is not particularly limited to this, and the output processing may be omitted. In addition, control device 50 uses the amount of powder dust; however, the configuration is not particularly limited to this, and may acquire the amount of waste materials and execute the control processing and the output processing based on the amount of waste materials. FIG. 10 is a diagram of correspondence relationship information including the correspondence relationship between the waste material image and the amount of waste materials. FIG. 11 is a diagram of correspondence relationship information including the correspondence relationship between the amount of waste materials and the dust-proof control amount. FIG. 12 is a flowchart illustrating an example of another dust-proof control output processing routine. In FIG. 12, the same steps as those in the dust-proof control output processing routine described above are denoted by the same reference numerals, and detailed descriptions thereof will be omitted. As illustrated in FIGS. 10 and 11, control device 50 may use the correspondence relationship in which the width or area of waste material 12 obtained from the captured image is associated with the amount of waste materials and the correspondence relationship information in which the amount of waste materials is associated with the control amount of dust-proof section 60. In addition, as illustrated in FIG. 12, control section 51 acquires the amount of waste materials in S300, acquires the control amount corresponding to the amount of waste materials in S310, and acquires the control amount or the accumulated operation amount according to the amount of waste materials in S320. Also in control device 50, since the information on the control and the maintenance of dust-proof section 60 is output according to the increase and decrease in the amount of waste materials, it is possible to execute more appropriate maintenance of the device.

[0052] In the above-described embodiment, although control section 51 obtains the amount of powder dust and the amount of waste materials from waste material 12 based on width W of waste material 12 included in the captured image, control section 51 may also obtain the amount of powder dust and the amount of waste materials from the area using width W. The area of waste material 12 may be acquired by multiplying width W of waste material 12 by a predetermined coefficient. In this case, the predetermined coefficient may be, for example, a predetermined

distance (predetermined intervals) in conveyance direction D. Alternatively, multiple widths W may be acquired at multiple positions, and the area of waste material **12** may be acquired based on a line connecting acquired multiple widths W and a distance in conveyance direction D of a region including multiple positions. Alternatively, as the area of waste material **12**, the area of waste material **12** in a predetermined region in the captured image may be acquired based on the captured image. In addition, in the above-described embodiment, the amount of powder dust or the amount of waste materials from waste material **12** is obtained based on width W of waste material **12**; however, the configuration is not particularly limited to this. Control section **51** may acquire the height information of waste material **12** from detection section **33** located upstream of removal section **34** and that detects the height of the waste material, and obtain the amount of waste materials and/or the amount of powder dust from the waste material based on the height information of the waste material. For example, the amount of waste materials **12** can be obtained by deriving the volume of waste material **12** using width W, height H, and conveyance amount C of waste material **12**. In control device **50**, the information regarding the amount of waste materials can be obtained by relatively easy processing using the height of waste material **12**. For example, detection section **33** may acquire a first distance from the detection section to the conveyance surface of conveyance device **22** and a second distance from the detection section to the upper portion of waste material **12**. The difference between the first distance and the second distance may be acquired as the height of waste material **12**.

[0053] In the embodiment described above, although dust-proof section **60** includes gas circulation section **61**, liquid circulation section **65**, and filter section **75**, one or more of these components may be omitted, or other components may be added. In addition, gas circulation section **61** may be applied to other configurations such as each slider and control device **50**, liquid circulation section **65** may be used for other configurations, and filter section **75** may be applied to other configurations.

[0054] In the above-described embodiment, in foreign substance removal device **30**, imaging section **32** is used to detect a foreign substance present in the waste material and acquire the information regarding the amount of waste materials **12**; however, the configuration is not particularly limited to this, and foreign substance removal device **30** may include multiple imaging sections. From the viewpoint of suppressing the complexity of the device, it is desirable not to include multiple imaging sections **32**.

[0055] In the above-described embodiment, the present disclosure is described as foreign substance removal device **30** including control device **50**; however, the configuration is not particularly limited to this, and the present disclosure may be applied to control device **50** itself, or may be applied to a control method.

[0056] Here, control device **50** of the present disclosure may be configured as follows. For example, a control device of the present disclosure is a control device used in a foreign substance removal device including a removal section configured to pick up and remove a foreign substance from a waste material conveyed by a conveyance device, a dust-proof section configured to perform dust-proofing related to maintenance of the foreign substance removal device with respect to powder dust derived from the waste material, and an imaging section located upstream of the removal section and configured to capture the image of the waste material, the control device including a control section configured to acquire an amount of powder dust as information regarding an amount of waste material from a captured image captured by the imaging section.

[0057] In this control device, the amount of powder dust is acquired as the information regarding the amount of waste material from the captured image obtained by capturing the image of the waste material. In this control device, since the amount of powder dust derived from the waste material can be obtained, it is possible to more appropriately execute the maintenance of the device when the foreign substance is removed from the waste material using the amount of powder dust.

[0058] A control method of the present disclosure is a control method used in a foreign substance

removal device including a removal section configured to pick up and remove a foreign substance from a waste material conveyed by a conveyance device, a dust-proof section configured to perform dust-proofing related to maintenance of the foreign substance removal device with respect to powder dust derived from the waste material, and an imaging section located upstream of the removal section and configured to capture an image of the waste material, the control method including a step of executing one or more of control processing of controlling the dust-proof section based on information regarding an amount of waste material obtained from a captured image captured by the imaging section, and output processing of outputting information regarding the maintenance of the foreign substance removal device based on the information regarding the amount of waste material obtained from the captured image captured by the imaging section.

[0059] In this control method, similar to the control device described above, since the information on the control and the maintenance of the dust-proof section is output according to the increase and decrease in the amount of waste material, it is possible to more appropriately execute the maintenance of the device. In the control method, various aspects of the above-described control device may be adopted, or steps for achieving each function of the above-described control device may be added.

[0060] Alternatively, a control method of the present disclosure is a control method used in a foreign substance removal device including a removal section configured to pick up and remove a foreign substance from a waste material conveyed by a conveyance device, a dust-proof section configured to perform dust-proofing related to maintenance of the foreign substance removal device with respect to powder dust derived from the waste material, and an imaging section located upstream of the removal section and configured to capture the image of the waste material and removing the foreign substance contained in the waste material, the control method including a step of acquiring an amount of powder dust as information regarding an amount of waste material from a captured image captured by the imaging section.

[0061] In this control method, similar to the control device described above, since the amount of powder dust derived from the waste material can be obtained, it is possible to more appropriately execute the maintenance of the device using the amount of powder dust. In the control method, various aspects of the above-described control device may be adopted, or steps for achieving each function of the above-described control device may be added.

INDUSTRIAL APPLICABILITY

[0062] The present disclosure can be used in technical fields such as recycling of industrial waste.

REFERENCE SIGNS LIST

[0063] **10**: Recycling system,

[0064] **12**: Waste material,

[0065] **13**: Target object,

[0066] **14**: Foreign substance,

[0067] **15**: Primary crusher,

[0068] **16**: Primary magnetic separator,

[0069] **17**: Screen machine,

[0070] **18**: Secondary crusher,

[0071] **19**: Secondary magnetic separator,

[0072] **20 to 25**: Conveyance device,

[0073] **30**: Foreign substance removal device,

[0074] **31**: Recognition section,

[0075] **32**: Imaging section,

[0076] **33**: Detection section,

[0077] **34**: Removal section,

[0078] **35**: X-axis slider,

[0079] **36**: Drive mechanism,

[0080] **37**: Screw shaft,
[0081] **38**: Nut,
[0082] **39**: Motor,
[0083] **40**: Y-axis slider,
[0084] **41**: Lifting and lowering section,
[0085] **42**: Drive mechanism,
[0086] **43**: Screw shaft,
[0087] **44**: Nut,
[0088] **45**: Motor,
[0089] **46**: Pickup section,
[0090] **48**: Dust-proof cover,
[0091] **50**: Control device,
[0092] **51**: Control section,
[0093] **52**: Storage section,
[0094] **53**: Accumulated information,
[0095] **54**: Threshold value information,
[0096] **55**: Correspondence relationship information,
[0097] **56**: Cooling fan,
[0098] **57**: Support portion,
[0099] **58**: Removal head,
[0100] **59**: Dust box,
[0101] **60**: Dust-proof section,
[0102] **61** Gas circulation section,
[0103] **62**: Air supply section,
[0104] **63**: Supply pipe,
[0105] **64**: Discharge port,
[0106] **65**: Liquid circulation section,
[0107] **66**: Slider dust-proof section,
[0108] **67**: Cleaning liquid tube,
[0109] **68**: Circulation pump,
[0110] **70**: Lifting and lowering dust-proof section,
[0111] **71**: Cleaning liquid tube,
[0112] **72**: Circulation pump,
[0113] **75**: Filter section,
[0114] **76**: Filter,
[0115] **D**: Conveyance direction,
[0116] **M**: Worker,
[0117] **W, W1, W2**: Width.

Claims

1. A control device used in a foreign substance removal device including a removal section configured to pick up and remove a foreign substance from a waste material conveyed by a conveyance device, a dust-proof section configured to perform dust-proofing related to maintenance of the foreign substance removal device with respect to powder dust derived from the waste material, and an imaging section located upstream of the removal section and configured to capture an image of the waste material, the control device comprising: a control section configured to execute one or more of control processing of controlling the dust-proof section based on information regarding an amount of waste material obtained from a captured image captured by the imaging section, and output processing of outputting information regarding the maintenance of the

foreign substance removal device based on the information regarding the amount of waste material obtained from the captured image captured by the imaging section.

2. The control device according to claim 1, wherein the control section is configured to acquire an amount of powder dust as the information regarding the amount of waste material from the captured image, and execute the control processing and/or the output processing based on the acquired amount of powder dust.

3. A control device used in a foreign substance removal device including a removal section configured to pick up and remove a foreign substance from a waste material conveyed by a conveyance device, a dust-proof section configured to perform dust-proofing related to maintenance of the foreign substance removal device with respect to powder dust derived from the waste material, and an imaging section located upstream of the removal section and configured to capture an image of the waste material, the control device comprising: a control section configured to acquire an amount of powder dust as information regarding an amount of waste material from a captured image captured by the imaging section.

4. The control device according to claim 3, wherein the control section is configured to execute one or more of control processing of controlling the dust-proof section based on the amount of powder dust and output processing of outputting information regarding the maintenance of the foreign substance removal device based on the amount of powder dust.

5. The control device according to claim 1, wherein the control section is configured to obtain the amount of waste material and/or an amount of powder dust from the waste material, based on information including at least a width of the waste material included in the captured image.

6. The control device according to claim 1, wherein the control section is configured to acquire height information of the waste material from a detection section, and obtain an amount of waste material and/or the amount of powder dust from the waste material, based on the height information of the waste material, the detection section being located upstream of the removal section and configured to detect a height of the waste material.

7. The control device according to claim 1, wherein the control section is configured to execute either control processing of controlling the dust-proof section with a tendency that the smaller the amount of waste material obtained from the captured image and/or an amount of powder dust from the waste material obtained from the captured image, the more an operation of the dust-proof section is suppressed, or output processing of outputting the information regarding the maintenance of the foreign substance removal device with a tendency that the smaller the amount of waste material obtained from the captured image and/or the amount of powder dust from the waste material obtained from the captured image, the longer an output interval of the information regarding the maintenance.

8. The control device according to claim 1, wherein the control section is configured to execute either control processing of controlling the dust-proof section with a tendency that the larger the amount of waste material obtained from the captured image and/or an amount of powder dust from the waste material obtained from the captured image, the more an operation of the dust-proof section is increased, or output processing of outputting the information regarding the maintenance of the foreign substance removal device with a tendency that the larger the amount of waste material obtained from the captured image and/or the amount of powder dust from the waste material obtained from the captured image, the shorter an output interval of the information regarding the maintenance.

9. The control device according to claim 1, wherein the control section is configured to execute the output processing of outputting the information regarding the maintenance of the foreign substance removal device based on the information regarding the amount of waste material obtained from the captured image captured by the imaging section, accumulate an operation amount of the dust-proof section in the output processing, and output the information regarding the maintenance when the accumulated operation amount reaches a predetermined maintenance execution amount.

10. The control device according to claim 1, wherein the dust-proof section includes one or more of a gas circulation section configured to cause a gas to flow to remove powder dust from the waste material, a liquid circulation section configured to cause a liquid to flow to remove powder dust from the waste material, and a filter section configured to filter powder dust from the waste material.

11. The control device according to claim 1, wherein the removal section is configured to remove a foreign substance present in the waste material based on the captured image captured by the imaging section.

12. A foreign substance removal device comprising: a removal section configured to pick up and remove a foreign substance from a waste material conveyed by a conveyance device; a dust-proof section configured to perform dust-proofing related to maintenance of the foreign substance removal device with respect to powder dust derived from the waste material; an imaging section located upstream of the removal section and configured to capture an image of the waste material; and the control device according to claim 1.

13. A control method used in a foreign substance removal device including a removal section configured to pick up and remove a foreign substance from a waste material conveyed by a conveyance device, a dust-proof section configured to perform dust-proofing related to maintenance of the foreign substance removal device with respect to powder dust derived from the waste material, and an imaging section located upstream of the removal section and configured to capture an image of the waste material, the control method comprising: a step of executing one or more of control processing of controlling the dust-proof section based on information regarding an amount of waste material obtained from a captured image captured by the imaging section, and output processing of outputting information regarding the maintenance of the foreign substance removal device based on the information regarding the amount of waste material obtained from the captured image captured by the imaging section.

14. A control method used in a foreign substance removal device including a removal section configured to pick up and remove a foreign substance from a waste material conveyed by a conveyance device, a dust-proof section configured to perform dust-proofing related to maintenance of the foreign substance removal device with respect to powder dust derived from the waste material, and an imaging section located upstream of the removal section and configured to capture an image of the waste material, the control method comprising: a step of acquiring an amount of powder dust as information regarding an amount of waste material from a captured image captured by the imaging section.
